README - PCLP3 Project - Part I (Classification)

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1. Problem Type

The selected problem is binary classification. The goal is to determine whether a patient is ill or healthy based on medical and behavioral characteristics. The dataset was generated synthetically, with controlled distributions and balanced classes.

2. Dataset Structure

The dataset contains 700 instances, split into two subsets:
- Training subset: 500 rows
- Test subset: 200 rows

Each row corresponds to an anonymous patient and describes values for several attributes.

3. Features

> age: Patient age (between 18 and 90)
> sex: Biological sex (Male/Female)
> smoking: Smoker (Yes/No)
> alcohol: Alcohol consumption (Yes/No)
> exercise_freq: Frequency of physical activity per week (0-6)
> blood_pressure: Systolic blood pressure (float)
> cholesterol: Total cholesterol level (float)
> blood_sugar: Blood sugar level (float)
> disease: Target variable: 1 = ill, 0 = healthy

4. Exploratory Data Analysis (EDA)

a) Missing Value Analysis

Based on the outputs below, we observe that the training set has between 4% and 5.8% missing values in most columns, whereas the percentage in the test set ranges from 3% to 7%. The target column (`disease`) has no missing values, which is essential for training the model.

Treatment strategy:
- Numerical variables: imputation using the respective mean
- Categorical variables: imputation using the most frequent value (mode)

```
TRAIN
Numar valori lipsa:
 age                23
sex                 21
smoking             28
alcohol             22
exercise_freq       27
blood_pressure      25
cholesterol         25
blood_sugar         29
disease              0
dtype: int64
Procent valori lipsa:
 age                4.6
sex                 4.2
smoking             5.6
alcohol             4.4
exercise_freq       5.4
blood_pressure      5.0
cholesterol         5.0
blood_sugar         5.8
disease             0.0
dtype: float64
```

```
TEST
Numar valori lipsa:
 age                 6
sex                  8
smoking              6
alcohol             10
exercise_freq       12
blood_pressure      11
cholesterol          7
blood_sugar         14
disease              0
dtype: int64
Procent valori lipsa:
 age                3.0
sex                 4.0
smoking             3.0
alcohol             5.0
exercise_freq       6.0
blood_pressure      5.5
cholesterol         3.5
blood_sugar         7.0
disease             0.0
dtype: float64
```

b) Descriptive Statistics
For the training subset:
- The mean age is around 50, indicating a balanced sample distribution.
- Blood sugar (`blood_sugar`) and cholesterol (`cholesterol`) have larger standard deviations, potentially indicating outliers or distinct patient groups.
- The categorical variables (`sex`, `smoking`, `alcohol`) are relatively evenly distributed, which is useful for the classification model.

b) Descriptive Statistics (continued)
Numerical and categorical descriptive statistics for the training and test sets are presented below.
Main observations:
- Mean age is very similar between training (52.88) and test (53.28), suggesting a stable sample.

- `blood_pressure`, `cholesterol`, and `blood_sugar` have high maximum values and large standard deviations, indicating greater dispersion and the possible presence of outliers.
- The distribution of categories (`sex`, `smoking`, `alcohol`) is relatively balanced in both training and test sets, with small acceptable variations that do not affect model learning.

```
Statistici numerice descriptive pentru Train:
                count        mean        std    min    25%    50%     75%    max
age             477.0   52.884696  21.046700   18.0   34.0   52.0   71.00   89.0
exercise_freq   473.0    2.968288   1.978981    0.0    1.0    3.0    5.00    6.0
blood_pressure  475.0  134.068000  25.786042   90.0  114.3  132.5  156.10  179.8
cholesterol     475.0  215.238737  51.863139  120.4  168.9  221.9  257.90  299.8
blood_sugar     471.0  131.179618  37.348819   70.0   99.9  127.0  162.75  199.9
disease         500.0    0.480000   0.500100    0.0    0.0    0.0    1.00    1.0

Statistici numerice descriptive pentru Test:
                count        mean        std    min      25%    50%    75%    max
age             194.0   53.283505  20.568778   18.0   35.000   52.0   71.0   89.0
exercise_freq   188.0    3.154255   1.968368    0.0    1.000    3.0    5.0    6.0
blood_pressure  189.0  135.890476  26.150081   90.1  116.000  136.9  159.0  179.4
cholesterol     193.0  205.736788  49.663161  120.3  163.600  199.9  245.5  299.5
blood_sugar     186.0  140.958602  37.388999   72.1  107.825  143.2  175.6  199.6
disease         200.0    0.485000   0.501029    0.0    0.000    0.0    1.0    1.0
```

```
Statistici categorice descriptive pentru Train:

sex:
sex
Female    244
Male      235
Name: count, dtype: int64

smoking:
smoking
Yes    250
No     222
Name: count, dtype: int64

alcohol:
alcohol
No     248
Yes    230
Name: count, dtype: int64
```

```
Statistici categorice descriptive pentru Test:

sex:
sex
Male      105
Female     87
Name: count, dtype: int64

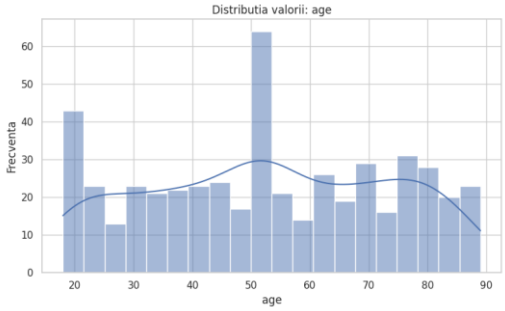
smoking:
smoking
No     103
Yes     91
Name: count, dtype: int64

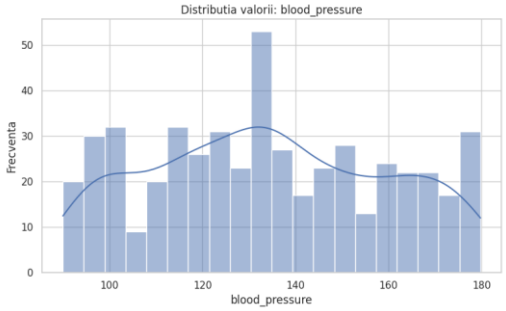
alcohol:
alcohol
No     99
Yes    91
Name: count, dtype: int64
```

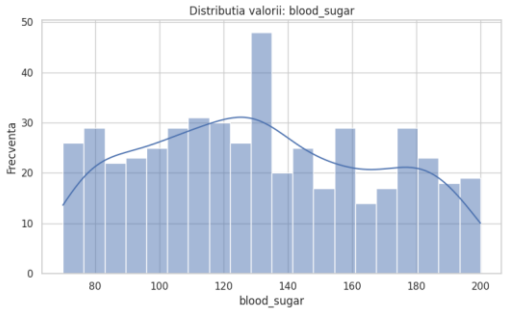
c) Variable Distribution Analysis
Distribution analysis helps us understand how the values of each numerical and categorical feature are distributed across the training and test sets. Histograms are used for numerical variables, and count plots for categorical variables.

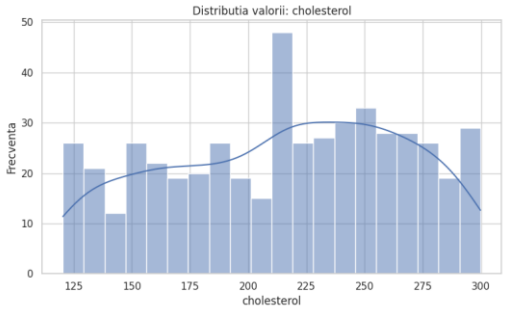
Observations:
- Histograms show a relatively uniform age distribution; cholesterol and blood sugar appear approximately normally distributed but with tails toward higher values (right-skewed).
- Count plots show that categorical variables are balanced between categories (e.g., Male/Female, Yes/No). This balance is favorable for model training.

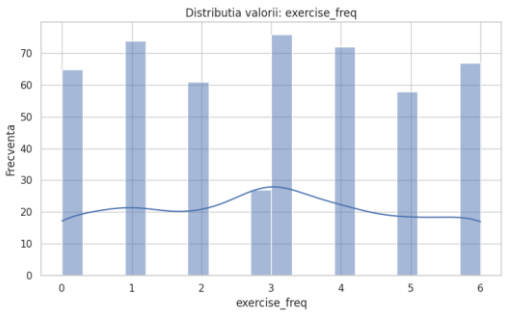
Histograms - Training



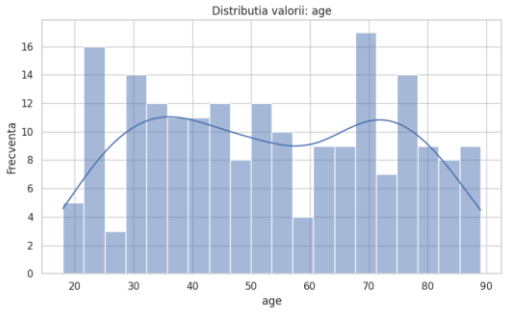


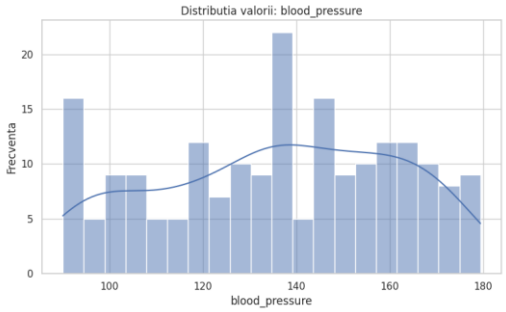


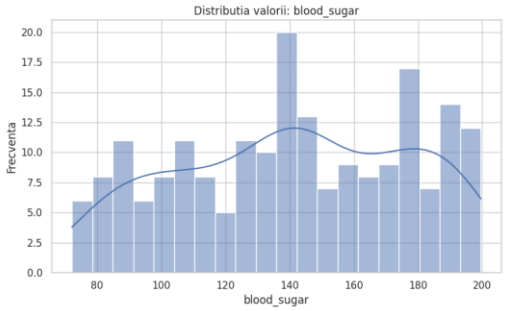


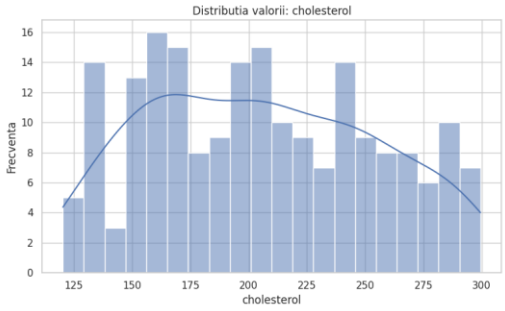


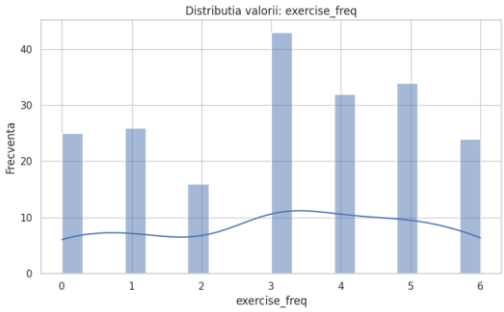
Histograms - Test



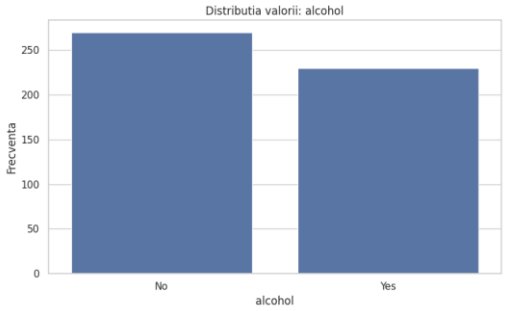


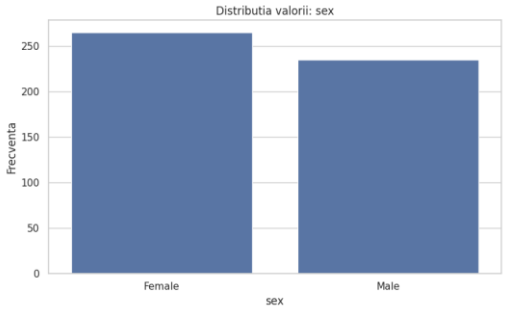


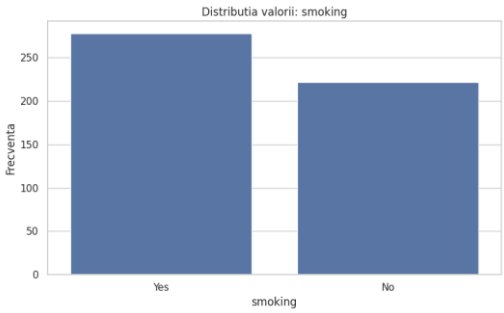




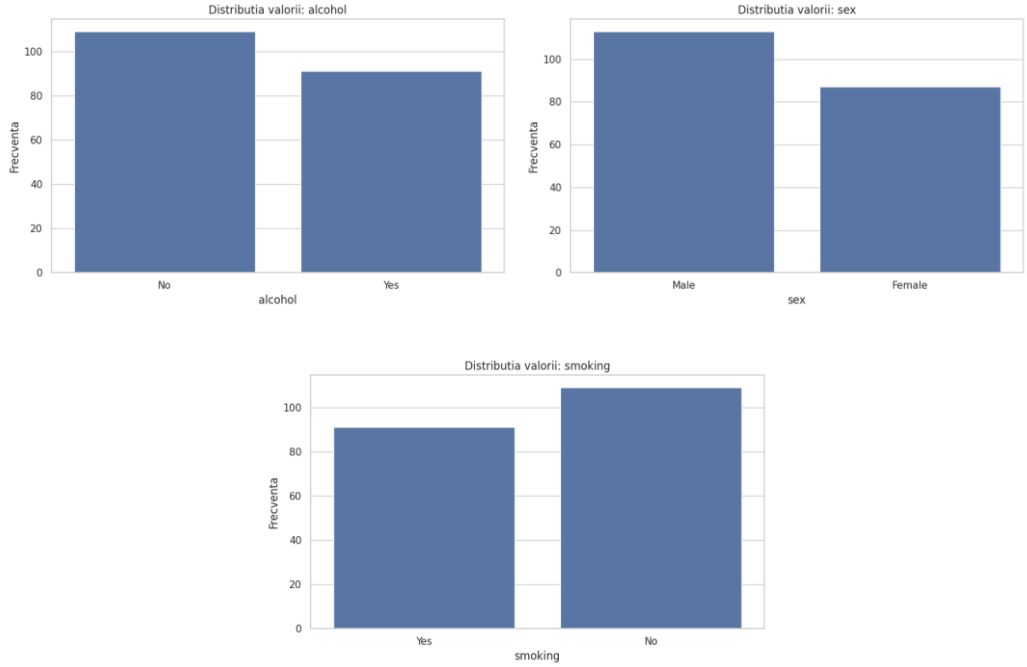
Count Plots - Training







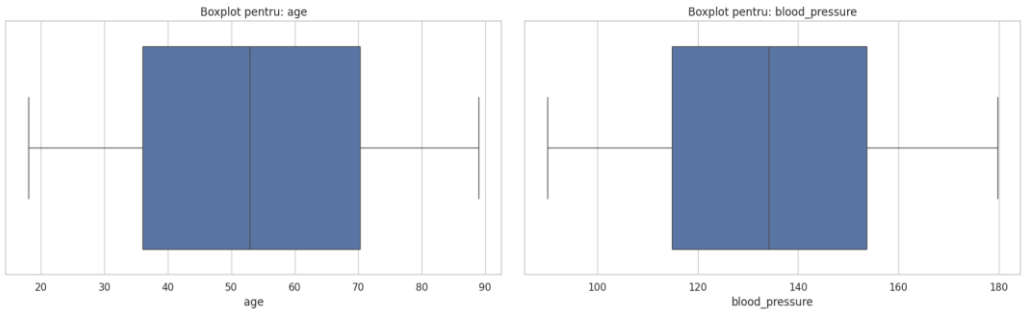
Count Plots - Test

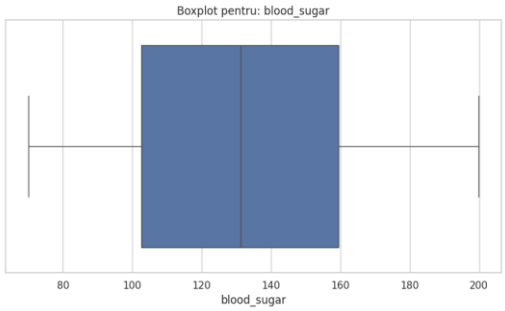


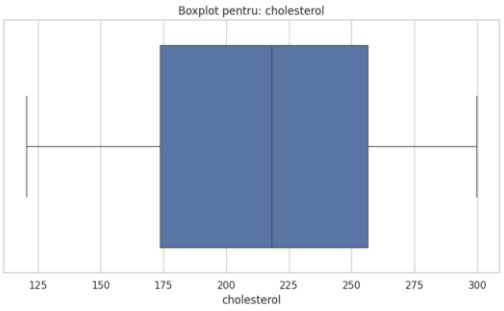
d) Outlier Detection

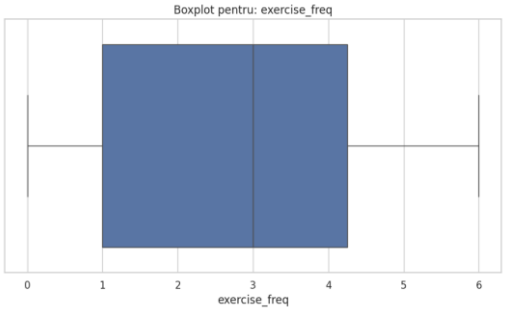
Box plots highlight extreme values that deviate from the main distribution of numerical features. In particular, high `cholesterol` and `blood_sugar` values stand out, lying significantly beyond the upper bounds of the distributions. These may represent patients with severe conditions.

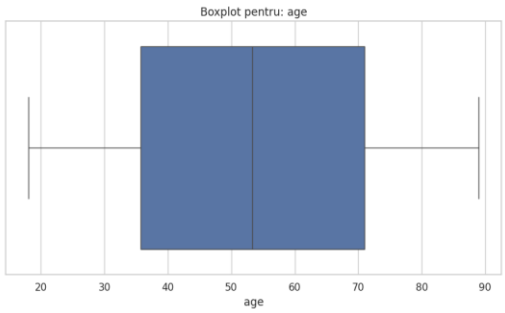
To prevent negative effects on the model, it is recommended either to remove outliers using the IQR rule or to apply a robust scaler that reduces the impact of extreme values.

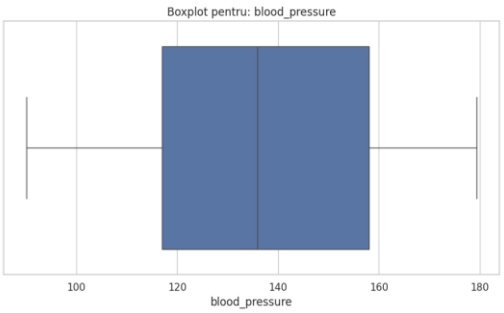


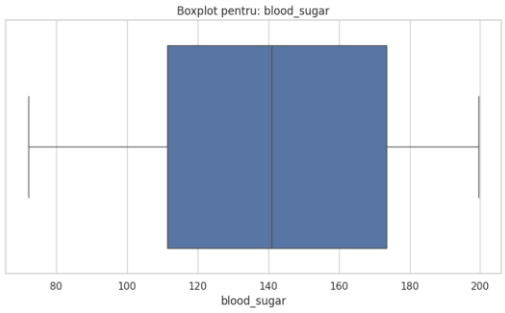

Boxplot pentru: blood_sugar
80
100
120
140
160
180
200
blood_sugar

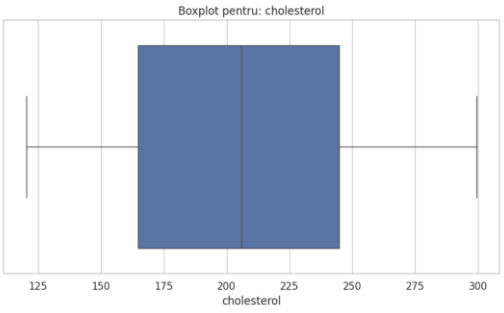

Boxplot pentru: cholesterol
125
150
175
200
225
250
275
300
cholesterol

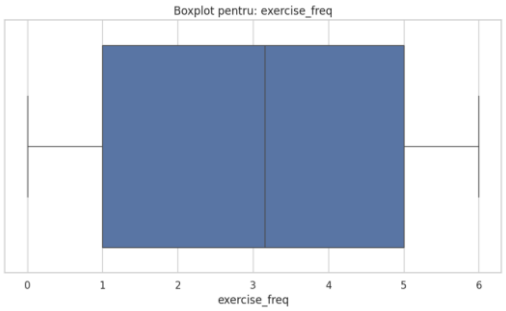

Boxplot pentru: exercise_freq
0
1
2
3
4
5
6
exercise_freq


Boxplot pentru: age
20
30
40
50
60
70
80
90
age


Boxplot pentru: blood_pressure
100
120
140
160
180
blood_pressure

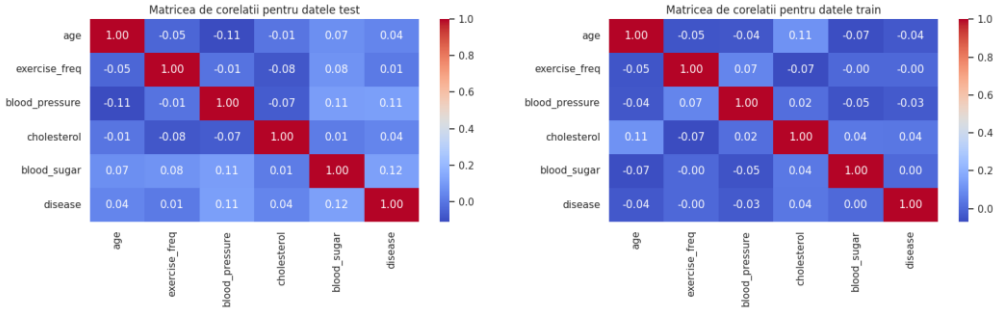

Boxplot pentru: blood_sugar
80
100
120
140
160
180
200
blood_sugar


Boxplot pentru: cholesterol
125
150
175
200
225
250
275
300
cholesterol



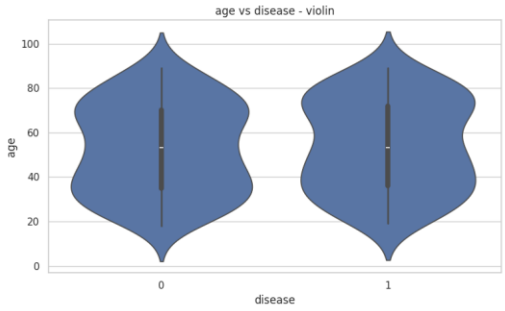
e) Correlation Analysis
Correlation heatmaps show linear relationships between numerical variables. A moderate correlation is observed between `blood_pressure` and `cholesterol`, suggesting these variables rise together. Both are also weakly positively correlated with `disease`, suggesting patients with higher values are more likely to be ill.

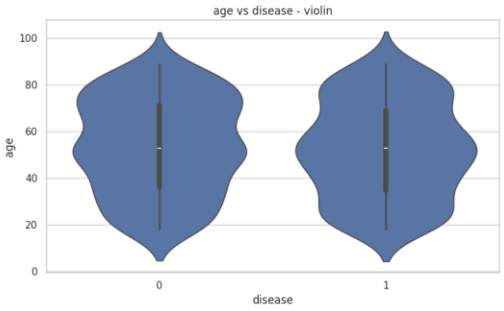
Because correlations are weak or moderate, no variable needs to be removed due to multicollinearity, but normalization could be helpful before training a logistic model.

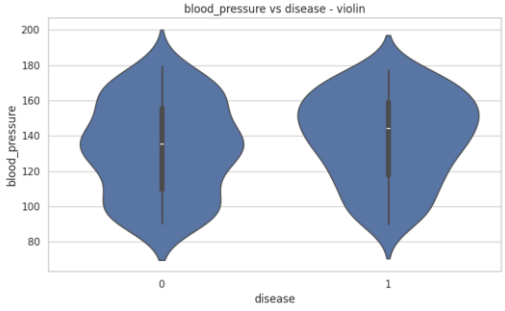


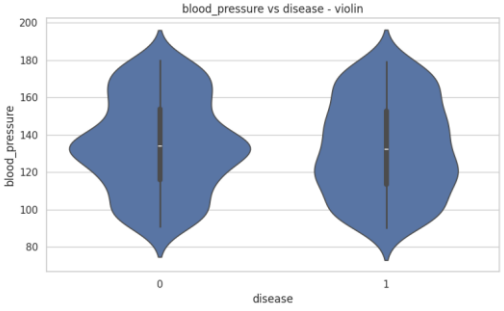
f) Relationship with the Target Variable (`disease`)
Violin and scatter plots show how each numerical feature varies according to the target variable (`disease`). A clear trend is observed: ill patients tend to have higher `blood_pressure` and `cholesterol` values. Distributions are wider and more dispersed for class 1 (ill), indicating greater variability.

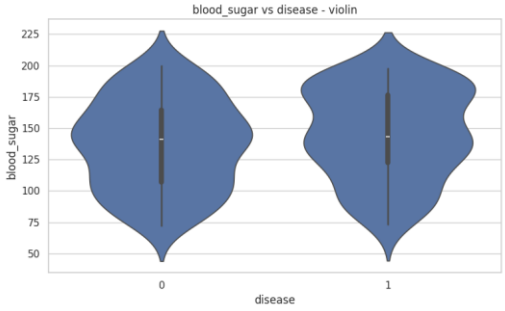
These features may help separate the classes, and some could benefit from transformations or combinations (e.g., a composite cholesterol-blood sugar score).

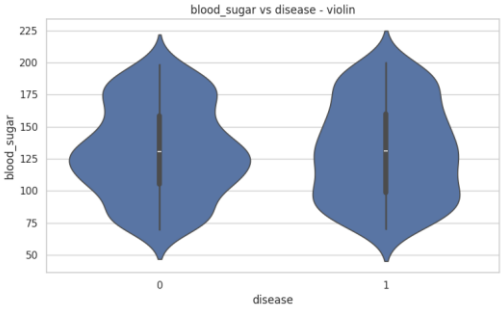

age vs disease - violin
age
disease
0
1

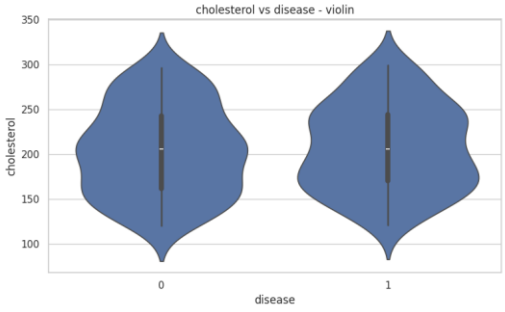

age vs disease - violin
age
disease
0
1

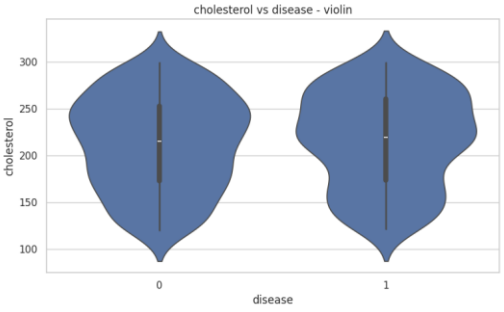

blood_pressure vs disease - violin
blood_pressure
disease
0
1

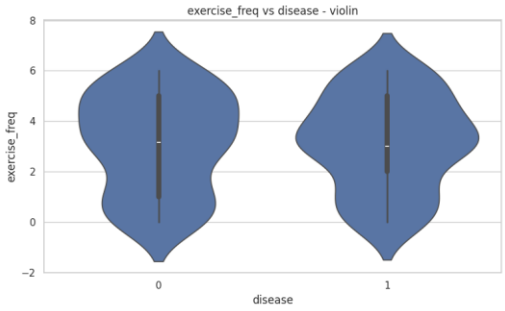

blood_pressure vs disease - violin
blood_pressure
disease
0
1

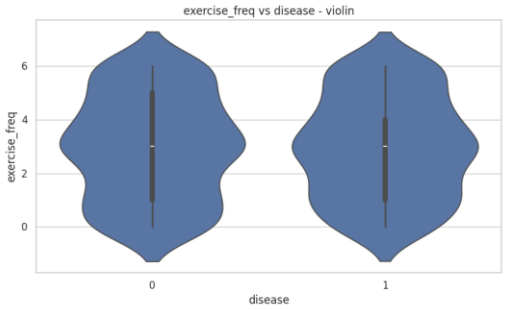

blood_sugar vs disease - violin
blood_sugar
disease
0
1

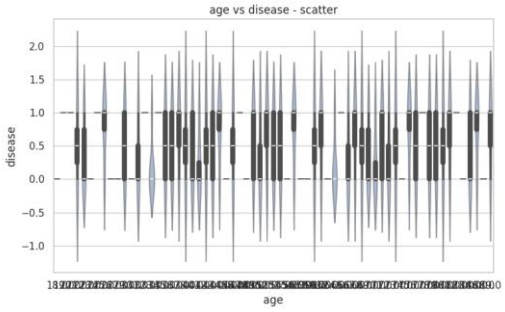

blood_sugar vs disease - violin
blood_sugar
disease
0
1

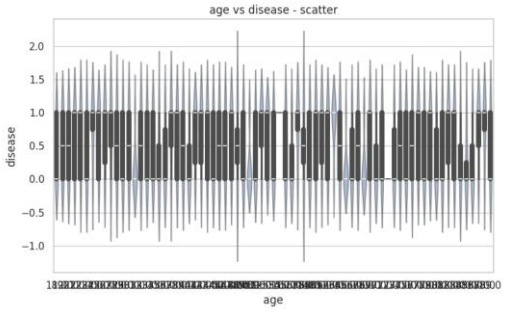

cholesterol vs disease - violin
cholesterol
disease
0
1

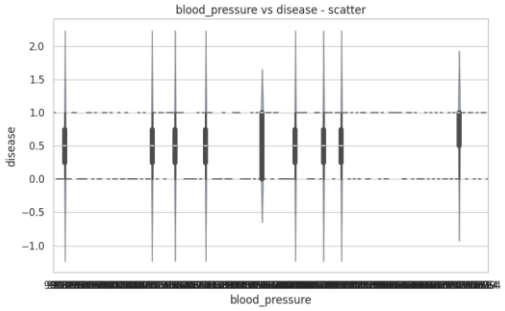

cholesterol vs disease - violin
cholesterol
disease
0
1

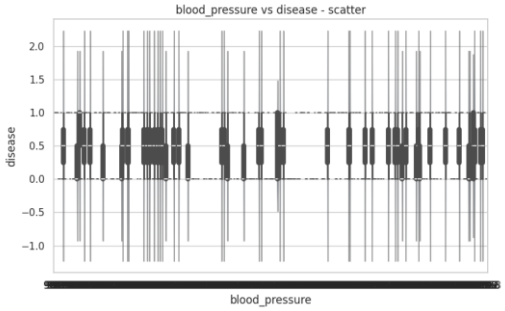

exercise_freq vs disease - violin
exercise_freq
disease
0
1

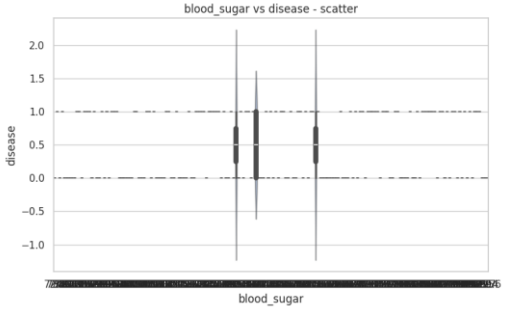

exercise_freq vs disease - violin
exercise_freq
disease
0
1

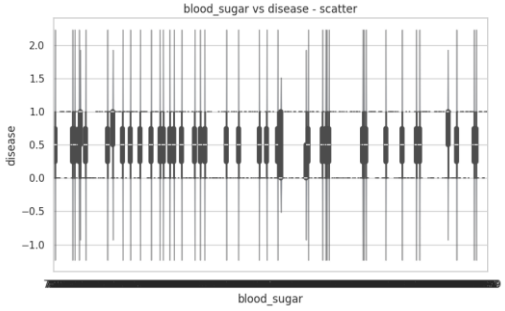

age vs disease - scatter
disease
age

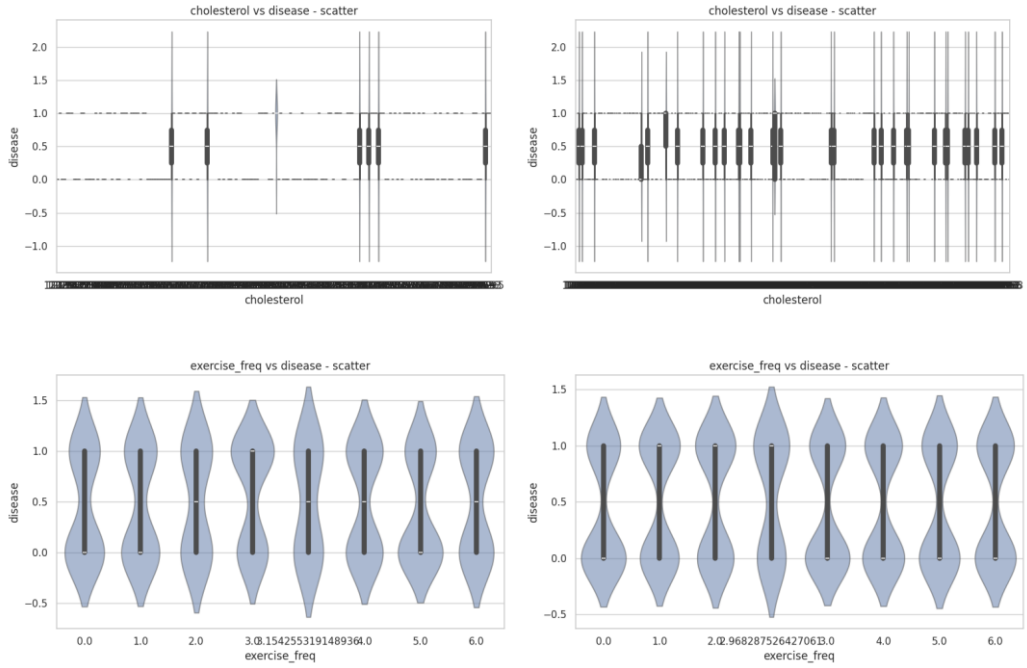

age vs disease - scatter
disease
age


blood_pressure vs disease - scatter
disease
blood_pressure


blood_pressure vs disease - scatter
disease
blood_pressure


blood_sugar vs disease - scatter
disease
blood_sugar


blood_sugar vs disease - scatter
disease
blood_sugar



5. Classification Model

For classification, a logistic regression model implemented with the `scikit-learn` library was used. This model was chosen because the problem involves binary classification and logistic regression offers interpretability and good performance for linear or mildly nonlinear data.

Before training, categorical variables (`sex`, `smoking`, `alcohol`) were encoded using one-hot encoding (`pd.get_dummies()` with `drop_first=True`). The `train` and `test` datasets were divided into features (`X_train`, `X_test`) and labels (`y_train`, `y_test`).

The model was trained on processed data, using 1,000 iterations to ensure convergence.

After training, the model was evaluated using the following metrics:
- Accuracy: proportion of correct predictions
- Precision: how many cases predicted as ill were classified correctly
- Recall: how many actually ill cases were correctly identified
- F1-score: harmonic mean of precision and recall

Evaluation was performed on the test set. The confusion matrix below shows the distribution of correctly and incorrectly predicted cases, divided into the healthy and ill classes.

